1. Difference Between OLTP and OLAP

OLTP: Manages daily business transactions, such as ATM withdrawals and online purchases.

OLAP: Analyzes large amounts of historical data to support business decisions.

Feature

OLTP

OLAP

Purpose

Daily transactions

Data analysis

Data

Current, detailed

Historical, summarized

Design

Normalized

Often denormalized

Operations

Insert, update, delete

Complex queries and analysis

Users

Cashiers, customers

Managers, analysts

Speed

Milliseconds

Seconds to minutes

Example

Banking transactions

Sales trend analysis

2. Difference Between OLTP and Data Warehousing

OLTP: Processes real-time business transactions.

Data Warehousing (DWH): Stores integrated historical data from multiple sources for reporting and analysis.

Feature

OLTP

Data Warehouse

Nature

Processing system

Data storage repository

Sources

Operational applications

Multiple data sources

Data

Frequently updated

Mostly historical and stable

Focus

Transaction speed

Analysis and reporting

Size

Usually smaller

Usually very large

Example

ATM system

Company-wide sales database

Conclusion

OLTP is used to manage daily business transactions quickly and accurately, while OLAP is used to analyze data for decision-making. A Data Warehouse stores large amounts of historical data collected from different sources. Together, these systems help organizations manage operations and make better business decisions.

References

Elmasri, R., & Navathe, S. B. Fundamentals of Database Systems. Pearson.

Silberschatz, A., Korth, H. F., & Sudarshan, S. Database System Concepts. McGraw Hill